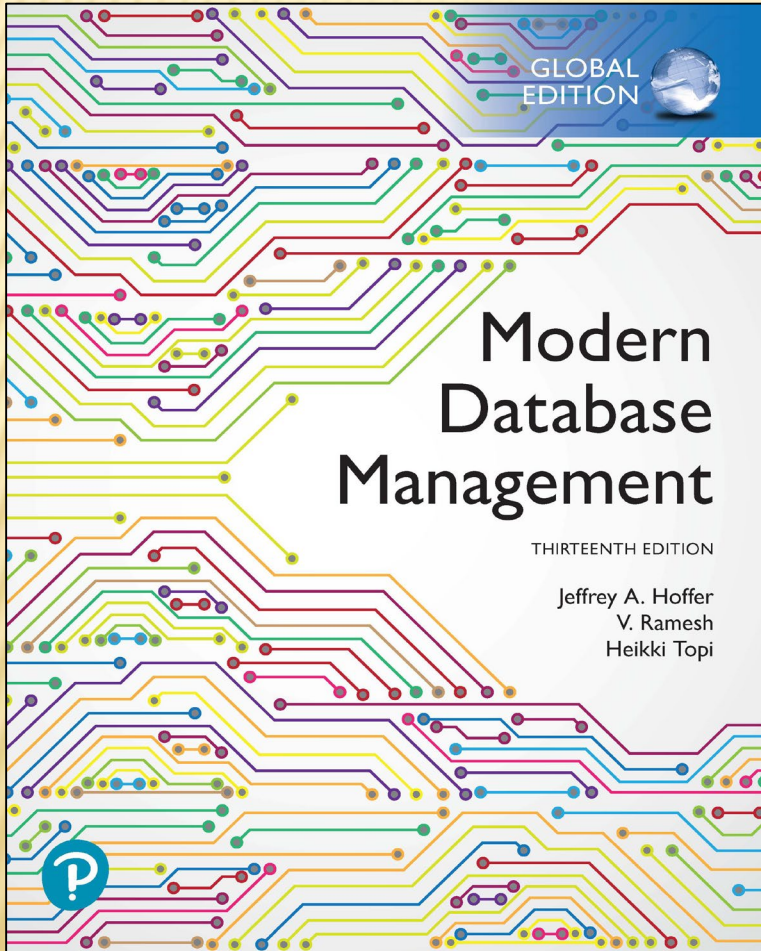


Modern Database Management

Thirteenth Edition



Chapter 1

The Database Environment and
Development Process

Learning Objectives (1 of 2)

1.1 Define terms

1.2 Name limitations of conventional file processing

1.3 Explain advantages of databases

1.4 Identify costs and risks of databases

1.5 Distinguish between operational and informational data management

1.6 List components of database environment

Learning Objectives (2 of 2)

- 1.7** Identify categories of database applications
- 1.8** Explain prototyping and agile development approaches
- 1.9** Explain roles of individuals
- 1.10** Explain differences between personal, multi-tiered, and enterprise data management
- 1.11** Explain three-schema architectures (external, conceptual, internal)

Definitions

- ✗ **Database:** organized collection of logically related data
- ✗ **Data:** stored representations of meaningful objects and events
 - + **Structured:** numbers, text, dates 結構化
 - + **Unstructured:** images, video, documents 非結構化
- ✗ **Information:** data processed to increase knowledge in the person using the data
- ✗ **Metadata:** data that describes the properties and context of user data

Figure 1-1 Converting Data To Information (1 OF 2)

(a) Data in context

Class Roster

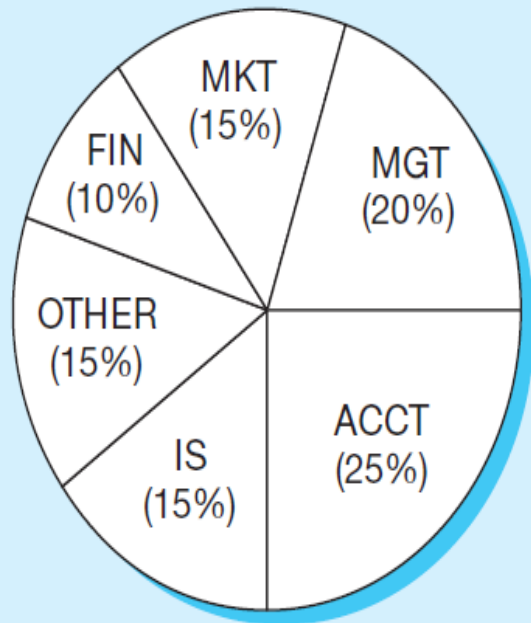
Course: MGT 500 Semester: Spring 2018
Business Policy

Section: 2

<u>Name</u>	<u>ID</u>	<u>Major</u>	<u>GPA</u>
Baker, Kenneth D.	324917628	MGT	2.9
Doyle, Joan E.	476193248	MKT	3.4
Finkle, Clive R.	548429344	PRM	2.8
Lewis, John C.	551742186	MGT	3.7
McFerran, Debra R.	409723145	IS	2.9
Sisneros, Michael	392416582	ACCT	3.3

Figure 1-1 Converting Data To Information (2 OF 2)

(b) Summarized data



Percent Enrollment by Major (2018)

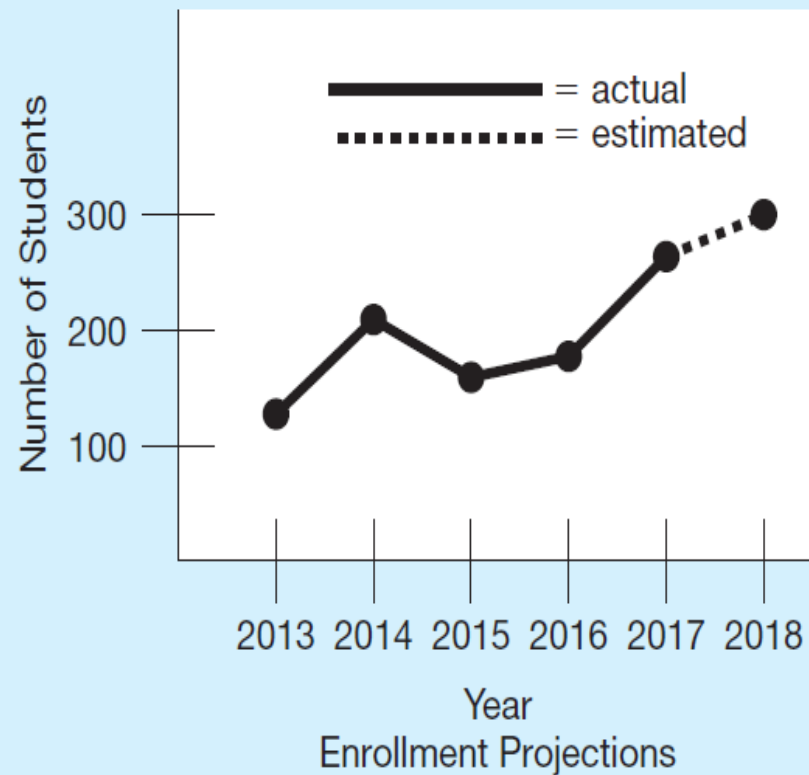


Table 1-1 Example Metadata For Class Roster

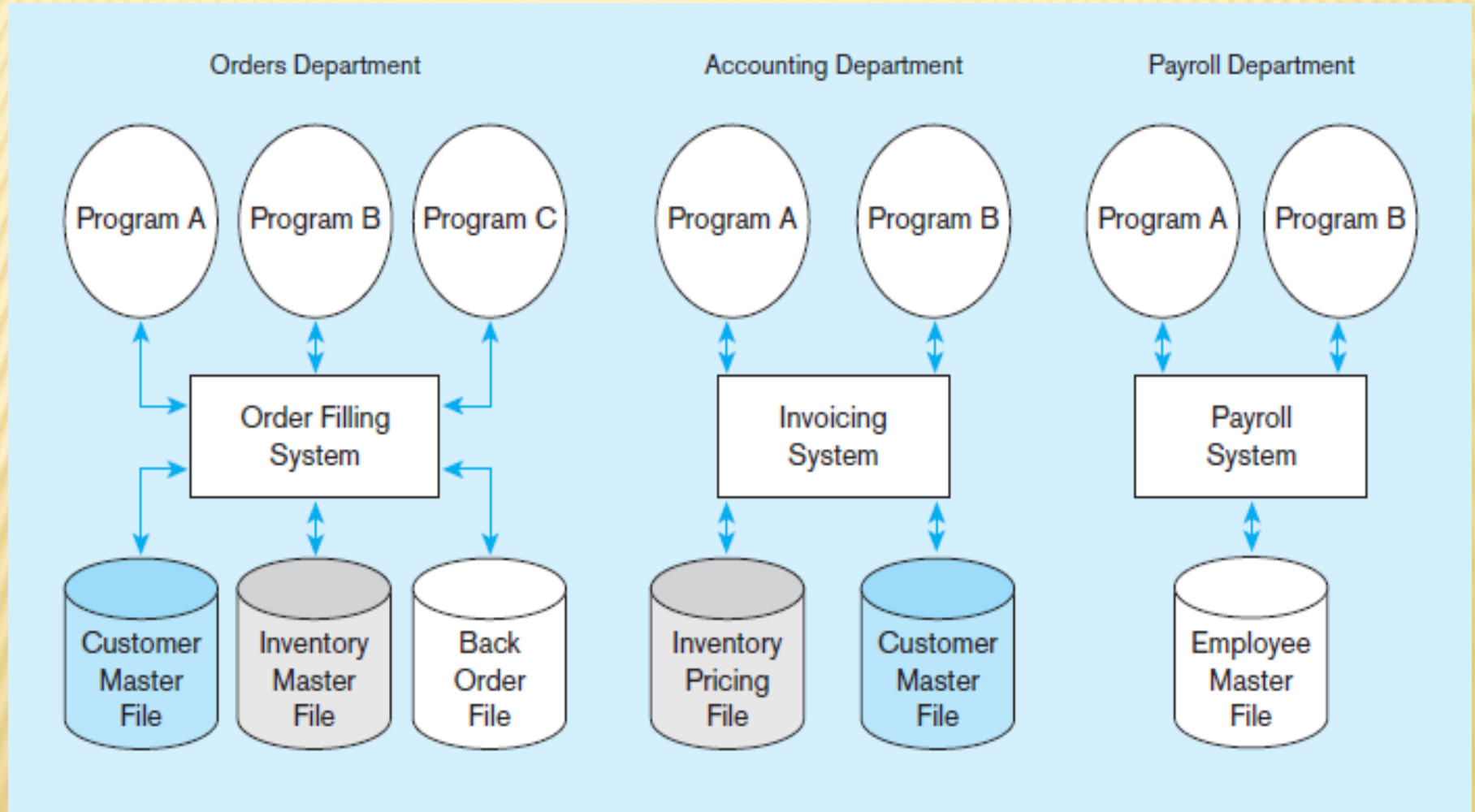
- ✗ Descriptions of the properties or characteristics of the data, including data types, field sizes, allowable values, and data context

Name	Type	Length	Min	Max	Description	Source
Course	Alphanumeric	30	Blank	Blank	Course ID and name	Academic Unit
Section	Integer	1	1	9	Section number	Registrar
Semester	Alphanumeric	10	Blank	Blank	Semester and year	Registrar
Name	Alphanumeric	30	Blank	Blank	Student name	Student IS
ID	Integer	9	Blank	Blank	Student ID (SSN)	Student IS
Major	Alphanumeric	4	Blank	Blank	Student major	Student IS
GPA	Decimal	3	0.0	4.0	Student grade point average	Academic Unit

Disadvantages Of File Processing

- ✗ Program-Data Dependence
 - + All programs maintain metadata for each file they use
- ✗ Duplication of Data
 - + Different systems/programs have separate copies of the same data
- ✗ Limited Data Sharing
 - + No centralized control of data
- ✗ Lengthy Development Times
 - + Programmers must design their own file formats
- ✗ Excessive Program Maintenance
 - + 80% of information systems budget

Figure 1-2 Old File Processing Systems at Pine Valley Furniture Company

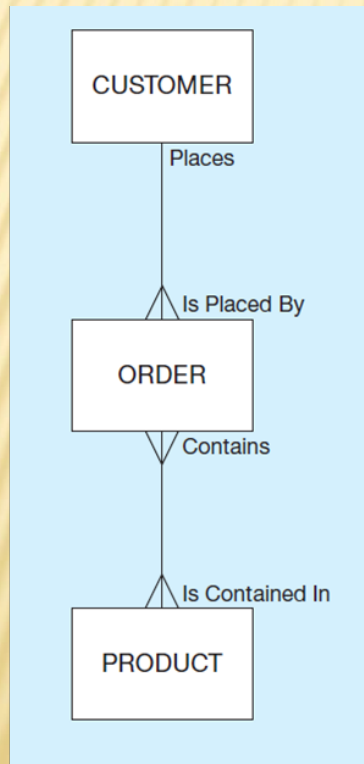


The Database Approach (1 of 2)

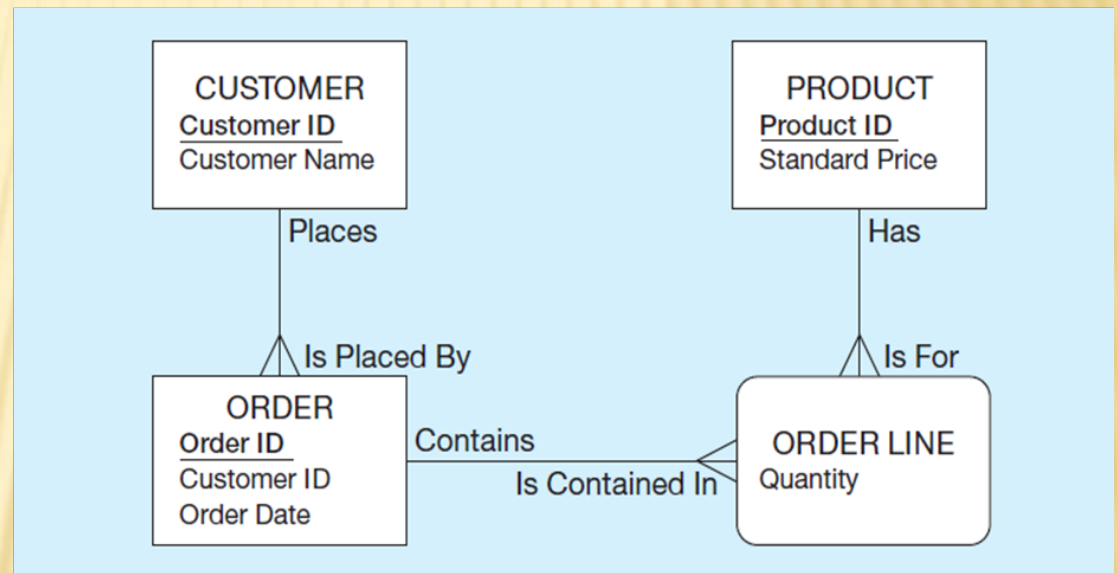
- Data models
 - Graphical diagram capturing nature and relationship of data
 - Enterprise Data Model – high-level entities and relationships for the organization
 - Project Data Model – more detailed view, matching data structure in database or data warehouse
 - **Entities** 實體
 - Noun form describing a person, place, object, event, or concept
 - Composed of **attributes** 屬性
 - **Relationships** 關聯
 - Between entities
- ER Model = Entity + Relationships**
- + Usually one-to-many (1:N) or many-to-many (M:N), but can also be one-to-one (1:1)

Figure 1-3 Comparison of Enterprise- and Project-Level Data Models

(a) Segment of an enterprise data model



(b) Segment of a project data model



The Database Approach (2 OF 2)

關聯表

- ✗ Relational Databases
- ✗ Database technology involving tables (relations) representing entities and **primary/foreign keys** representing relationships (see Figure 1-17)

主鍵

外來鍵

The screenshot displays four database tables in a software interface:

- Order_T**: Contains columns OrderID, OrderDate, and CustomerID. It lists 10 records with OrderID ranging from 1001 to 1010.
- Customer_T**: Contains columns CustomerID and CustomerName. It lists 15 records with CustomerID ranging from 1 to 15.
- OrderLine_T**: Contains columns OrderID, ProductID, and OrderedQuantity. It lists 18 records showing the quantity of each product ordered for each order.
- Product_T**: Contains columns ProductID, ProductDescription, and ProductFinish. It lists 8 records with ProductID ranging from 1 to 8.

Relationships are indicated by the presence of primary keys (OrderID in Order_T, CustomerID in Customer_T, ProductID in Product_T) and foreign keys (CustomerID in Order_T, OrderID in OrderLine_T, ProductID in OrderLine_T).

Advantages Of The Database Approach

- ✗ Program-data independence
- ✗ Planned data redundancy
- ✗ Improved data consistency
- ✗ Improved data sharing
- ✗ Increased application development productivity
- ✗ Enforcement of standards
- ✗ Improved data quality
- ✗ Improved data accessibility and responsiveness
- ✗ Reduced program maintenance
- ✗ Improved decision support

Costs And Risks Of The Database Approach

- ✖ New, specialized personnel
- ✖ Installation and management cost and complexity
- ✖ Conversion costs
- ✖ Need for explicit backup and recovery
- ✖ Organizational conflict

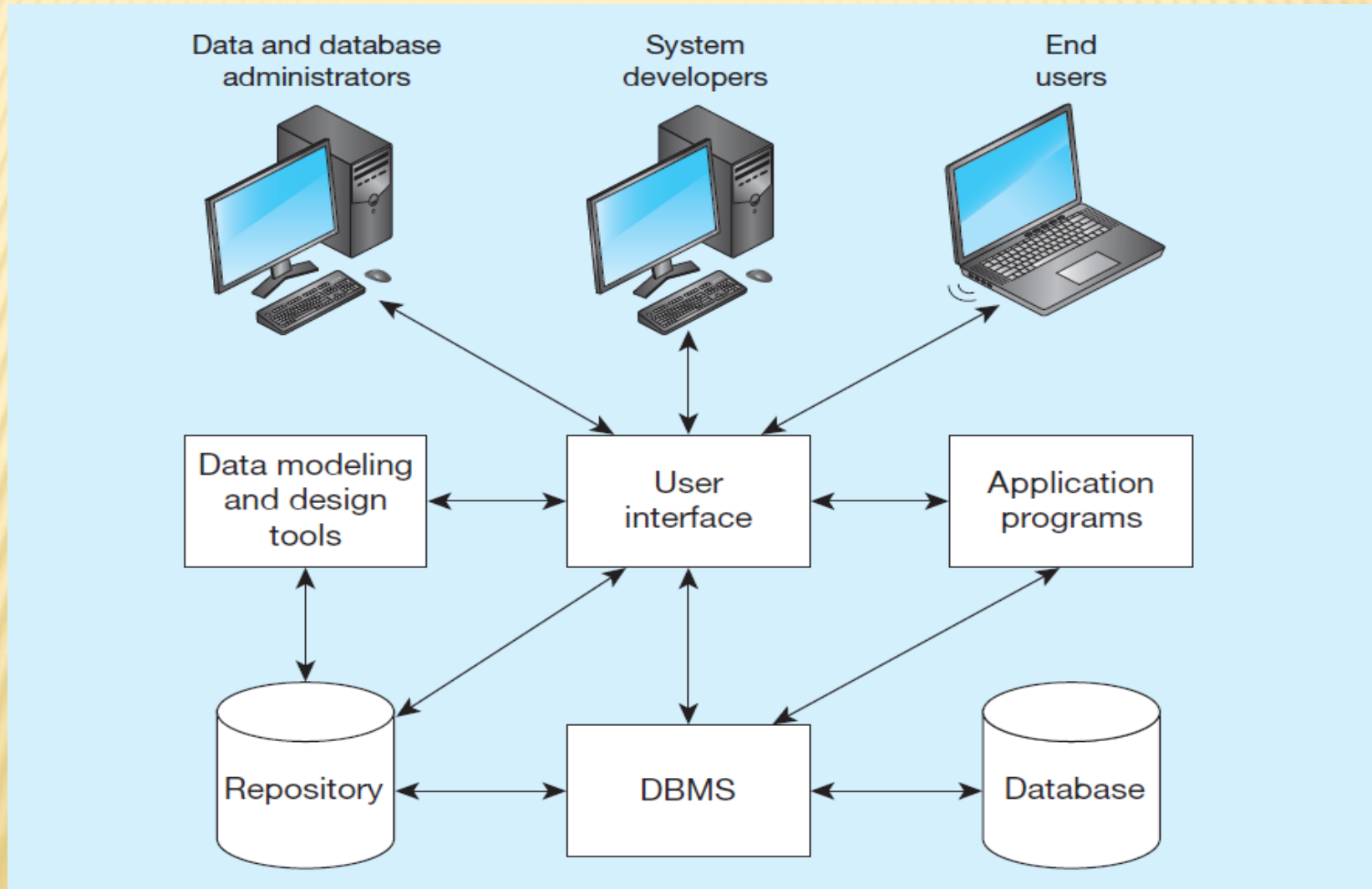
Figure 1-5 Integrated Data Management Framework

	Operational	Informational	
	Transactional	Analytical– Data Warehousing	Analytical– Big Data
Technology	Relational	Relational	Non-relational
Modeling	Conceptual data modeling with (E)ER (Chapters 2 and 3)		
Design	Logical data modeling with the relational model; Normalization (Chapter 4)		
Infrastructure	Physical design of relational databases; Security; Cloud computing (Chapter 8)	Data warehousing and data integration (Chapter 9)	Big data technologies, including Hadoop & NoSQL (Chapter 10)
Access	SQL (Chapters 5 and 6) Applications with SQL (Chapter 7)		
Data analysis	Analytics and its implications (Chapter 11)		
Governance and data management	Lifecycle (Chapter 1) Governance, data quality, and master data management (Chapter 12)		

Components Of The Database Environment

- ✗ Data modeling and design tools – automated tools used to design databases and application programs
- ✗ Repository – centralized storehouse of metadata
- ✗ **Database Management System (DBMS)** – software for managing the database
Access 最小 DBMS Oracle 最大型 DBMS
MS SQL 中型 DBMS
- ✗ Database – storehouse of the data
- ✗ Application Programs – software using the data
- ✗ User Interface – text, graphical displays, menus, etc. for user
- ✗ Data/Database Administrators – personnel responsible for maintaining the database
資料庫管理師
- ✗ System Developers – personnel responsible for designing databases and software
- ✗ End Users – people who use the applications and databases

Figure 1-6 Components of the Database Environment



The Database Development Process

✗ S D L C

- + System Development Life Cycle
- + Detailed, well-planned development process
- + Time-consuming, but comprehensive
- + Long development cycle

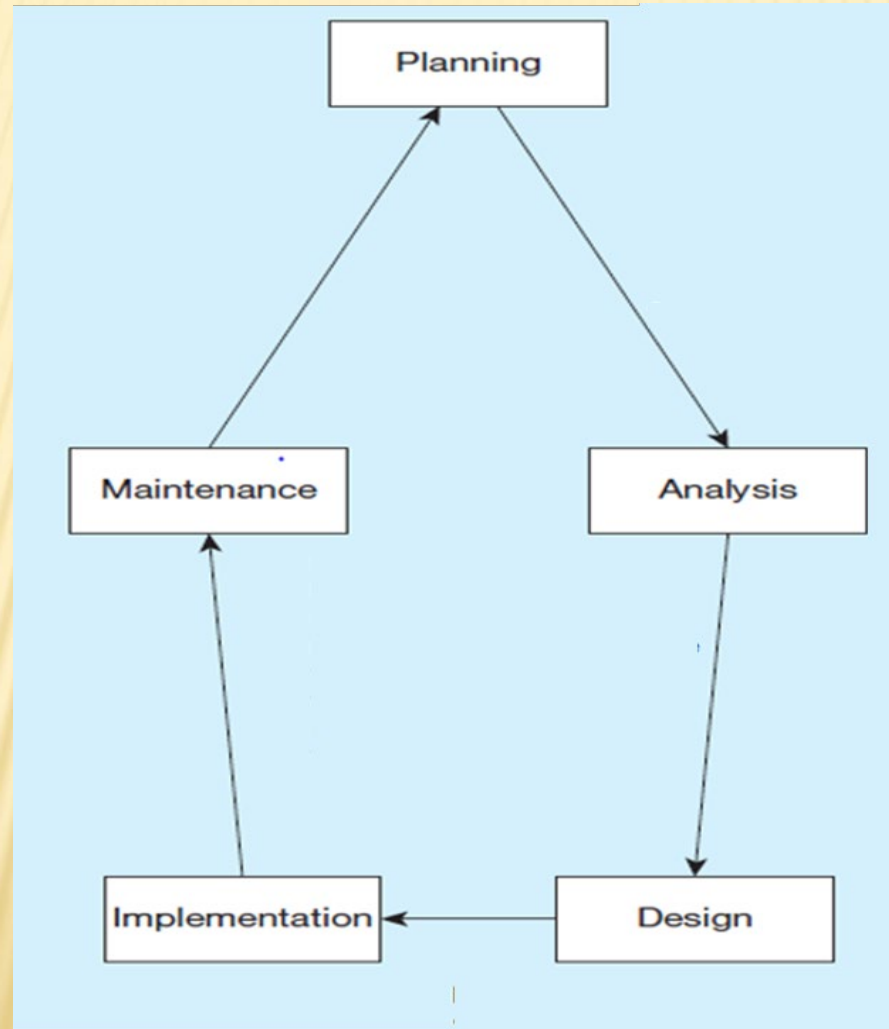
✗ Prototyping

- + Rapid application development (R A D)
- + cursory attempt at conceptual data modeling
- + Define database during development of initial prototype
- + Repeat implementation and maintenance activities with new prototype versions

Systems Development Life Cycle (SDLC)

- ✗ The traditional methodology used to develop, maintain, and replace information systems
- ✗ Five main steps:
 1. **Planning** – preliminary understanding of business situation. Enterprise model and conceptual data modeling.
 2. **Analysis** – thorough analysis of business situation, leading to functional requirements. Detailed conceptual data modeling.
 3. **Design** – logical and physical database design, to develop technology and organization.
 4. **Implementation** – writing programs, building databases, testing, installing, training, and documenting.
 5. **Maintenance** – monitoring, repairing, and enhancing.

From Figure 1-8 Database Development Activities During the SDLC



Alternative Information Systems Development Approaches

- ✖ Traditional SDLC: methodical, structured, and time consuming
- ✖ Rapid Application Development (R A D): faster and more adaptive, especially when a database is already in place
- ✖ Several flavors:
 - + Prototyping
 - + Agile methodologies
 - + eXtreme programming
 - + Scrum
 - + DSDM (dynamic system development methodologies)

Figure 1-9 The Prototyping Methodology and Database Development Process

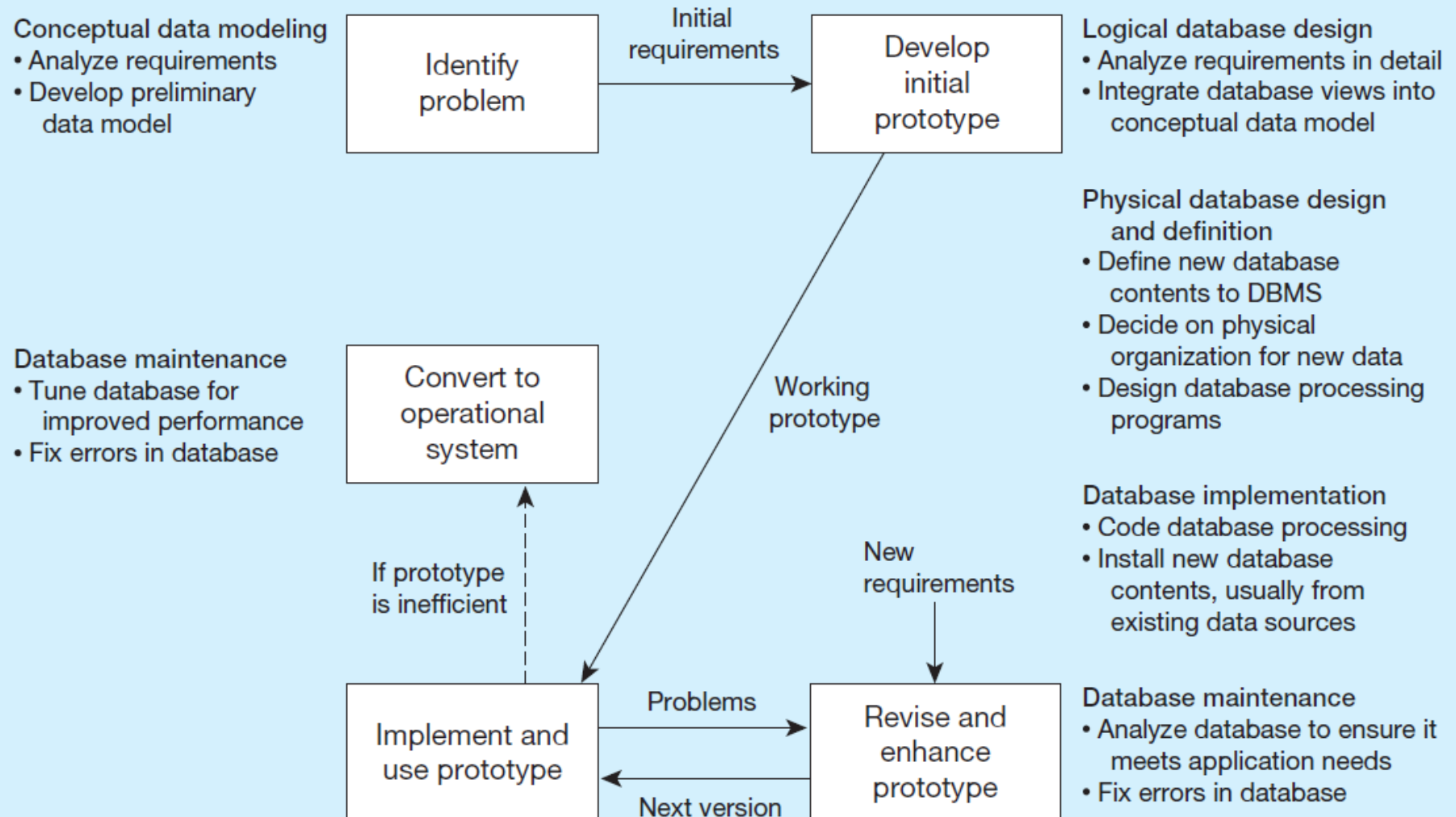
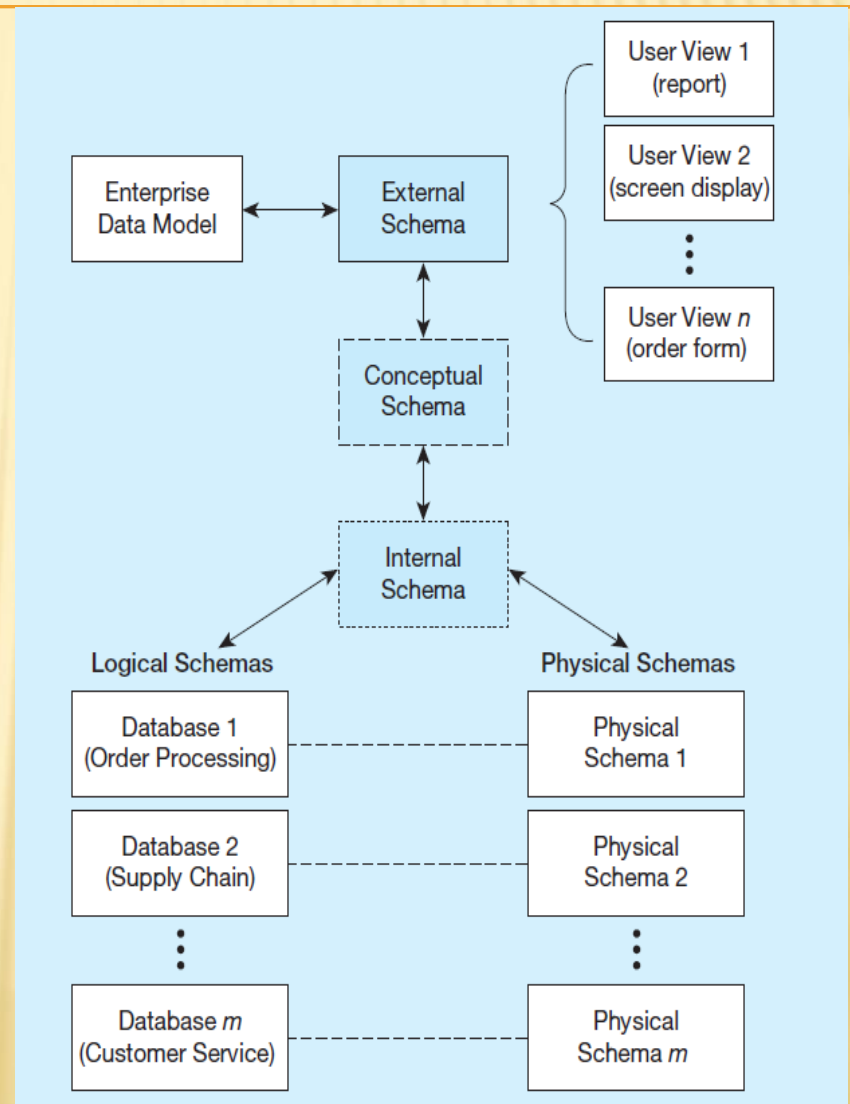


Figure 1-10 Three-schema Architecture

- ✗ Different people have different views of the database
- ✗ External Schema
 - + User Views
 - + Subsets of Conceptual Schema
 - + Can be determined from business-function/data entity matrices
 - + D B A determines schema for different users
- ✗ Conceptual Schema (E R models)
- ✗ Internal Schema (logical and physical structures)



Managing People Involved In Database Development

- ✖ Project – a planned undertaking of related activities to reach an objective that has a beginning and an end
- ✖ Initiated and planned in planning stage of SDLC
- ✖ Executed during analysis, design, and implementation
- ✖ Closed at the end of implementation

Database Project Team Members (1 OF 2)

- ✖ Business analysts – analyze business situation and establish requirements
- ✖ Systems analysts – like business analysts, but also have technical expertise for overall information systems
- ✖ Database analysts and data modelers – analysts who focus on database
- ✖ Users – the “customers” communicate their needs to analysts
- ✖ Programmers – coders of the programs that interact with the database

Database Project Team Members (2 OF 2)

- ✖ Database architects – establish standards for data in business units
- ✖ Data administrators – responsible for existing databases, ensuring data integrity and consistency
- ✖ Project managers – oversee the projects, manage the personnel
- ✖ Other technical experts – network, operating system, documentation, etc.

Evolution Of Database Systems

- ✕ Driven by four main objectives:
 - + Need for program-data independence in order to reduce maintenance
 - + Desire to manage more complex data types and structures
 - + Ease of data access for less technical personnel
 - + Need for more powerful decision support platforms

Figure 1-11 The Range Of Database Technologies: Past And Present (1 OF 4)

(a) Evolution of database technologies

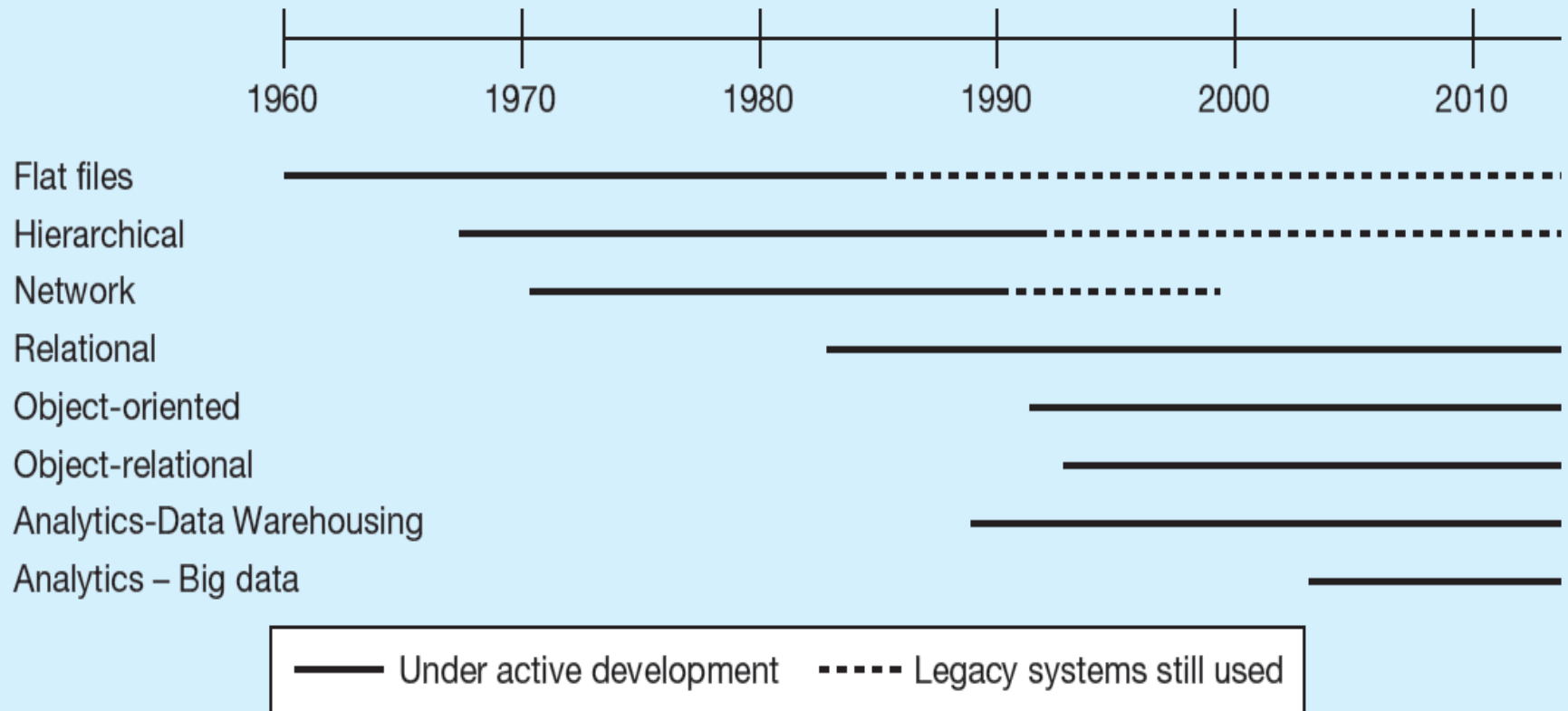


Figure 1-11 The Range Of Database Technologies: Past And Present (2 OF 4)

(b) Database architectures

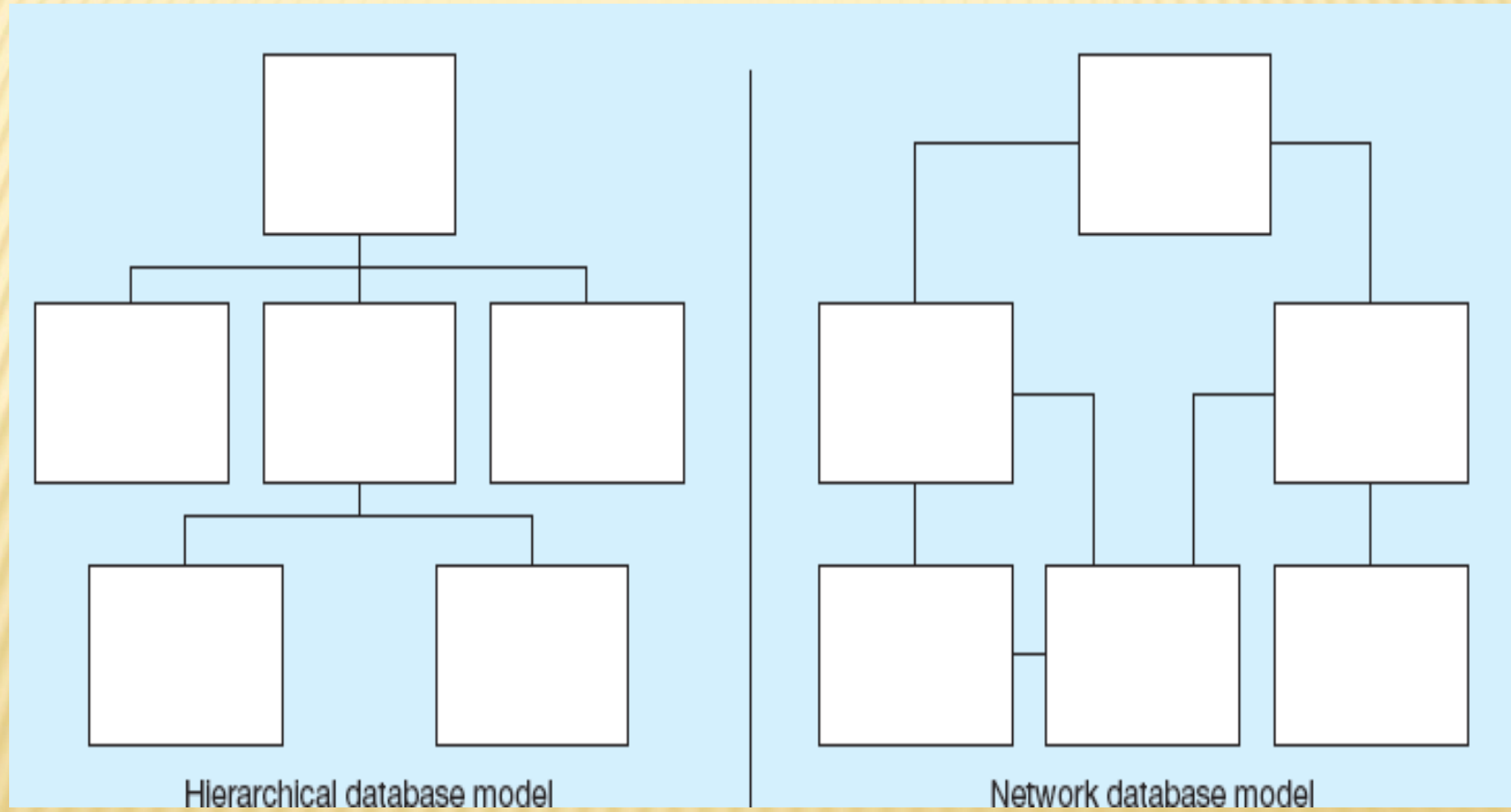


Figure 1-11 The Range Of Database Technologies: Past And Present (3 OF 4)

(b) Database architectures

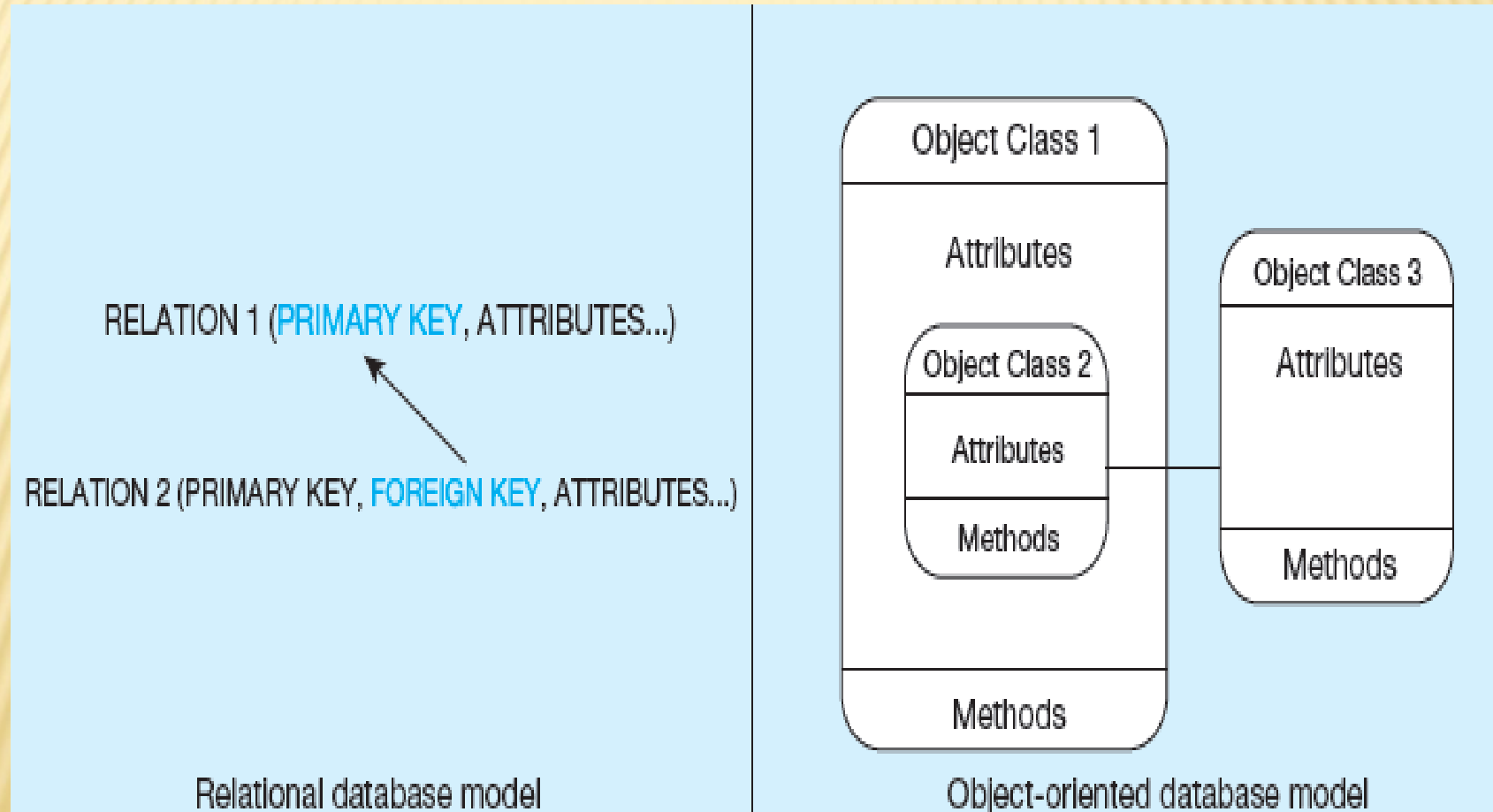
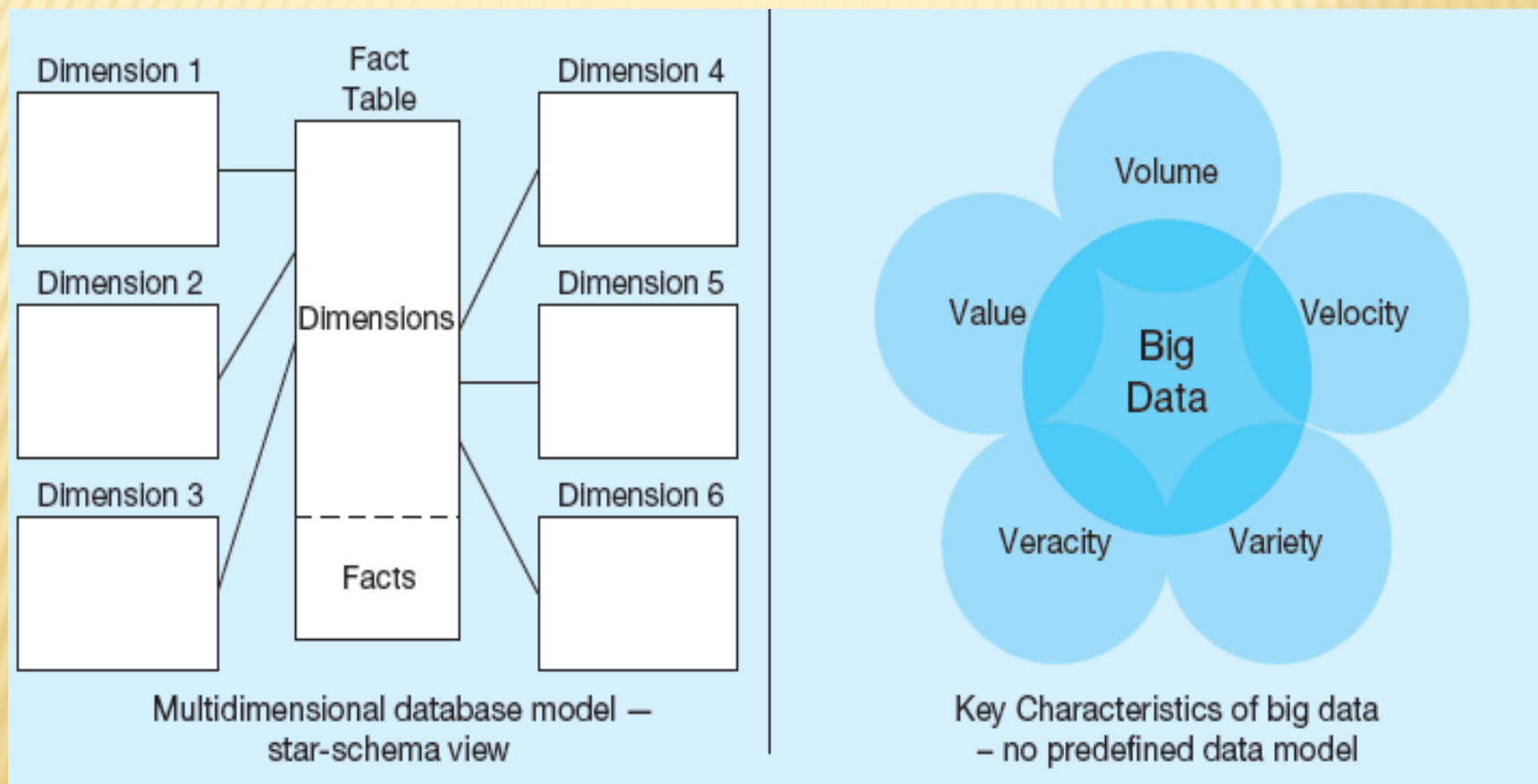


Figure 1-11 The Range Of Database Technologies: Past And Present (4 Of 4)

(b) Database architectures



The Range Of Database Applications

- ✗ Personal Databases

- + Typical size in the megabytes
- + Intended for one user

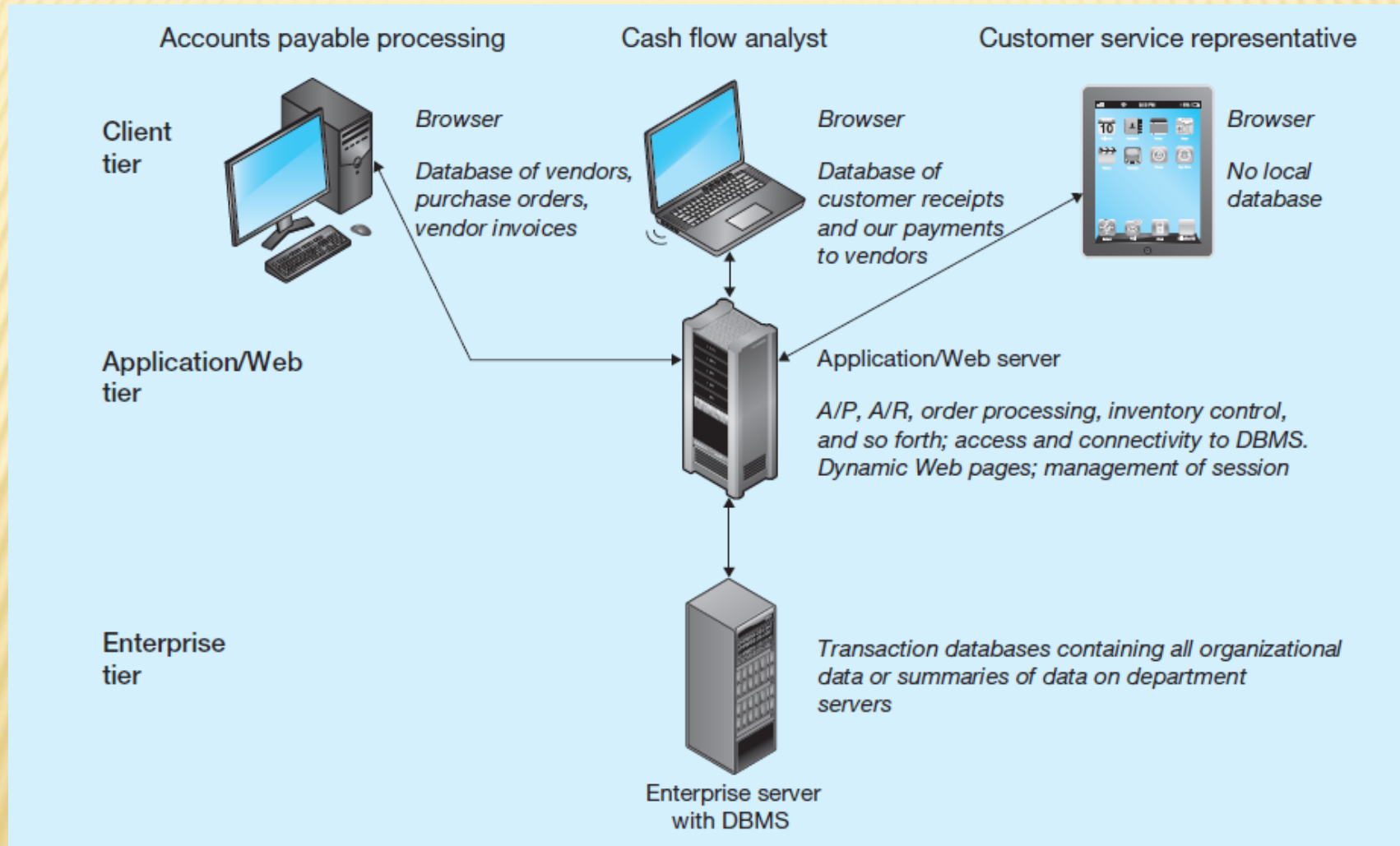
- ✗ Departmental Multi-Tiered Client/Server Databases

- + Typical size in the gigabytes
- + Intended for several users, usually doesn't exceed 100, department wide

- ✗ Enterprise Applications

- + Typical size in the gigabytes, terabytes, or even petabytes
- + Intended for a very large user base, company wide

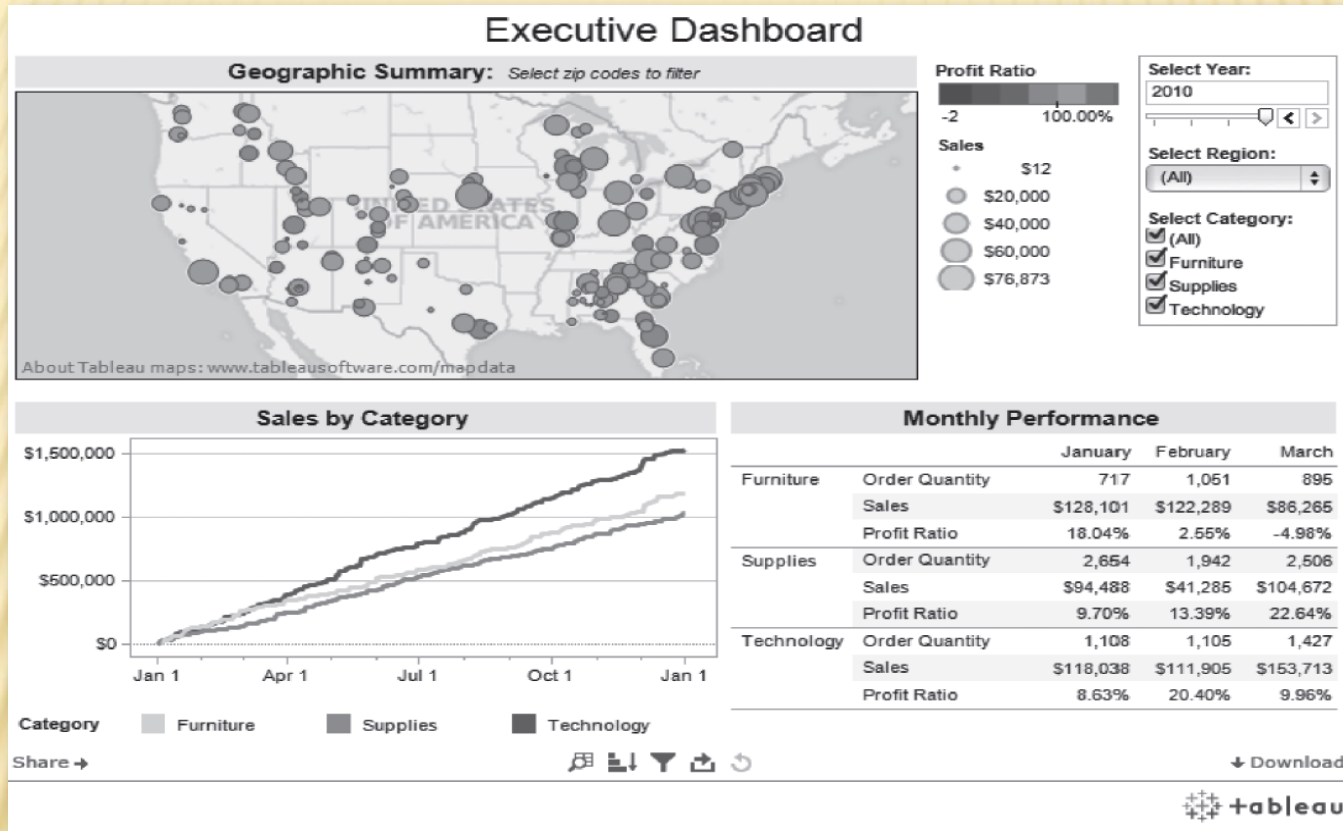
Figure 1-12 Multi-Tiered Client/Server Database Architecture



Types Of Enterprise Applications

- Enterprise Systems (typically involve relational databases)
 - + Backbone of an organization
 - + Enterprise resource planning (E R P)
 - + Customer relationship management
 - + Supply chain management
 - + Human resource management and payroll
- Data Warehouses (typically involve relational databases)
 - + Integrates data from multiple data sources
 - + Maintain historical data
 - + Help identify patterns and trends
- ✗ Data Lakes (often don't involve relational databases)
 - + Large integrated repository for internal and external data that does not follow a predefined schema

Figure 1-13 An Example of an Executive Dashboard



(http://public.tableausoftware.com/profile/mirandali#!/vizhome/Executive-Dashboard_7/ExecutiveDashboard)

Courtesy Tableau Software

PINE VALLEY FURNITURE (PVF)

- ✖ PVF is a fictional furniture company
- ✖ Will be used as a case throughout the book
- ✖ Chapter 1 topics include
 - + Database Evolution at Pine Valley Furniture Company
 - + Project planning
 - + Analyzing database requirements
 - + Designing the database
 - + Using the database
 - + Administering the database
 - + Future of databases at PVF